Huawei Access Node ELEC-ACC-57 QoS/Congestion Event Troubleshooting SOP

Node: ELEC-ACC-57 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_4472

Description

The Node ID ELEC-ACC-57, located at the Electronic City Hub site, is experiencing a QoS/congestion event indicated by Alarm Code ALARM_4472 of severity Warning. This failure domain arises due to the occurrence of microbursts and interface tail drops, ultimately leading to QoS queue starvation. In Huawei's Access equipment, this phenomenon is precipitated by the interaction between the QoS policy and the interface's bandwidth limitation. When the interface's bandwidth is exceeded, it triggers the tail drop mechanism, causing QoS queues to become congested and subsequently resulting in microbursts. Furthermore, the Huawei Access node's scheduling mechanism, which prioritizes packets based on their QoS markings, can exacerbate this issue by allocating excessive bandwidth to high-priority packets, further degrading network performance. Consequently, it is essential to investigate and resolve this QoS/congestion event to maintain optimal network functionality.

Pre-requisite Checks

- Check: Verify that redundant links are functioning correctly and have sufficient bandwidth to handle traffic, Status: Complete

- Check: Confirm the Electronic City Hub site's environmental conditions are within normal operating parameters, Status: Complete

- Check: Dispatch remote hands to the Electronic City Hub site to access the Node ID ELEC-ACC-57, Status: Complete

Potential Root Causes

Cause	Probability
QoS policy misconfiguration	High
Interface bandwidth limitation	Medium
Scheduling mechanism inefficiency	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy, Expected Output: Policy name:
qos-policy-1, Description: This command displays the configured QoS
policy, allowing us to identify any misconfigurations or
inconsistencies that may be contributing to the QoS/congestion event.

Command: display interface brief, Expected Output: Interface Name IP
Address Status MTU, Description: This command provides a brief
overview of the interface's status, including its IP address, status,
and MTU, which can help us identify any issues related to interface
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```
bandwidth limitation.

Command: display scheduling, Expected Output: Scheduling name:
scheduling-1, Description: This command displays the scheduling
mechanism's configuration, enabling us to assess its efficiency and
potential impact on network performance.
```

Resolution Steps

Step 1: Step: Update QoS policy configuration, Command: qos policy update, Description: Update the QoS policy configuration to optimize packet prioritization and avoid excessive bandwidth allocation to high-priority packets.

Step 2: Step: Adjust interface bandwidth, Command: interface bandwidth adjust, Description: Adjust the interface's bandwidth to accommodate increased traffic demands and prevent tail drops.

Step 3: Step: Optimize scheduling mechanism, Command: scheduling optimize, Description: Optimize the scheduling mechanism to prioritize packets efficiently and prevent QoS queue starvation.

Post-Check Verification

```
Command: display qos policy, Expected Output: Policy name:
qos-policy-1 (updated), Description: Verify that the QoS policy
configuration has been updated correctly.

Command: display interface brief, Expected Output: Interface Name IP
Address Status MTU (adjusted), Description: Verify that the
interface's bandwidth has been adjusted correctly.

Command: display scheduling, Expected Output: Scheduling name:
scheduling-1 (optimized), Description: Verify that the scheduling
mechanism has been optimized correctly.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: qos policy restore, Description: Restore the original QoS policy configuration to its previous state if the updated configuration does not resolve the issue.

- Step: Revert interface bandwidth adjustment, Command: interface bandwidth revert, Description: Revert the interface's bandwidth to its original state if the adjustment does not resolve the issue.

- Step: Revert scheduling mechanism optimization, Command: scheduling revert, Description: Revert the scheduling mechanism to its original state if the optimization does not resolve the issue.